Optimal Execution

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Abstract

We aim to extract the optimal trading size when liquidating inventory given the current state of the orderbook affecting the execution of liquidating asset from inventory. We experiment to find exactly how to obtain the optimal trade size at each given time with the least execution cost and minimal reaction impact to the orderbook.

Keywords: Numerical Methods, Stochastic Calculus, Timeseries, Orderbook, Market-Making, Dynamic Programming, Hamilton-Jacobi-Bellman Equation

Motivation

Given a large block of inventory that needs to be liquidated over a period of time, we need a process that will sell the inventory without moving the market too much or taking on unnecessary price risk. The execution steps must take into account the balancing of the two opposing forces of market impact cost and timing risk.

Introduction

The Almgren–Chriss model (Almgren & Chriss,) constitutes a seminal quantitative framework that yields a closed-form solution for optimal execution trajectories that balance key market impact parameters in the liquidation of large asset positions via the limit order book. The resulting optimal trajectory can be used as a benchmark for more sophisticated dynamic execution algorithms.

Within this context, the execution of large orders is formulated as a stochastic control problem, in which the objective is to minimize a cost functional incorporating temporary price impact, permanent price impact, and execution (or market) risk. Under a specified set of modeling assumptions, these components are integrated into an analytically tractable optimization problem, from which a closed-form solution for the optimal liquidation schedule is derived.

Numerical Approximation

The objective of the liquidation problem is to find the optimal trading rate that minimizes the expected cost over a time horizon.

$$dx_s = v_s dx_s$$

where

- x_s: State variable representing the inventory level
- v_s: Control function of the trading rate or size of the inventory to trade per time step.
- dx_s: Change in inventory over time controlled by v_s.

The instantaneous cost for a given control v_s and state x_s is

$$\mathcal{L}(x_s, v_s) = \eta v_s^2 + \gamma\sigma^2 x_s^2$$

where:

- $\mathcal{L}(x_s, v_s)$: Instantaneous cost
- ηv^2: Cost related to the intensity of trade
- $\gamma\sigma^2 x^2$: Cost of inventory risk

The objective is to minimize the expected total cost from time t to T, starting from state x at time t.

$$V(x,t) = \min_{v_s, s\in[t,T]} \mathbb{E}\left[\int_t^T (\eta v^2 + \gamma^2 x^2) ds\right]$$

The expectation is taken over any potential random disturbances in the system. The state dynamics $dV = vdt$ is deterministic, and the expectation is often implicit.

Instead of solving the entire integral from time t to time T, we consider taking a small step Δt. Following Bellman's equation, the optimization problem is broken down into smaller problems. As such, the total cost of $V(t,x)$ is expressed as the cost incurred during the shorter interval $[t, t+\Delta t]$ plus the minimum cost from $[t+\Delta t]$ onward starting from the new state $V_{t+\Delta t}$.

$$\begin{aligned} V(x,t) = \min_{v_s, s\in[t,T]} \mathbb{E}\Bigg[&\int_t^{t+\Delta t} (\eta v_s^2 + \gamma\sigma^2 x_s^2) ds \\ &+ \int_{t+\Delta t}^T (\eta v_s^2 + \gamma\sigma^2 x_s^2) ds \Bigg] \end{aligned}$$

We approximate the first integral over the small interval $[t, t+\Delta t]$ by its value at time t

$$\int_t^{t+\Delta t} (\eta v_s^2 + \gamma\sigma^2 x_s^2) ds \approx (\eta v_t^2 + \gamma\sigma^2 x_t^2)\Delta t$$

And the state evolves deterministically:

$$x_{t+\Delta t} \approx x_t + v_t \Delta t$$

The second integral term is simply the optimal cost starting from time $t+\Delta t$ with state $x_{t+\Delta t}$, which is $V(t+\Delta t, x_{t+\Delta t})$.

The Bellman equation (Authors,) in discrete-step form is

$$V(t,x) = \min_{v_t} \left[\left(\eta v_t^2 + \sigma^2 x_t^2\right)\Delta t + V\left(t+\Delta t, x_t + v_t \Delta t\right) \right] \tag{1}$$

Now, we expand $V(t+\Delta t, x_t + v_t\Delta t)$ using Taylor series expansion around (t, x_t)

$$\begin{aligned} V(t+\Delta t, x_t + v_t\Delta t) \approx &V(t,x) + \frac{\partial V}{\partial t}(t,x)\Delta t + \frac{\partial V}{\partial x}(t,x)(v_t \Delta t) \\ &+ \frac{1}{2}\frac{\partial^2 V}{\partial t^2}(t,x)(\Delta t)^2 + \dots \end{aligned} \tag{2}$$

For small Δt, we can ignore terms of order $(\Delta t)^2$ and higher.

$$V(t+\Delta t, x_t + v_t\Delta t) \approx V(t,x) + \frac{\partial V}{\partial t}\Delta t + \frac{\partial V}{\partial x} v_t \Delta t$$

Substitute this into the Bellman equation in 1

$$V(t,x) \approx \min_{v_t} \left\{ (\eta v_t^2 + \gamma\sigma^2 x_t^2)\Delta t + V(t,x) + \frac{\partial V}{\partial t}\Delta t + \frac{\partial V}{\partial x} v_t \Delta t \right\}$$

Rearranged to isolate terms involving $V(t,x)$:

$$V(t,x) - V(t,x) \approx \min_{v_t} \left\{ \frac{\partial V}{\partial t}\Delta t + (\eta v_t^2 + \gamma\sigma^2 x_t^2)\Delta t + \frac{\partial V}{\partial x} v_t \Delta t \right\}$$

$$0 \approx \min_{v_t} \left\{ \frac{\partial V}{\partial t}\Delta t + \eta v_t^2 \Delta t + \gamma\sigma^2 x_t^2 \Delta t + \frac{\partial V}{\partial x} v_t \Delta t \right\}$$

Divide by Δt (assuming $\Delta t > 0$):

$$0 \approx \min_{v_t} \left\{ \frac{\partial V}{\partial t} + \eta v_t^2 + \gamma\sigma^2 x_t^2 + \frac{\partial V}{\partial x} v_t \right\}$$

This is the continuous-time Hamilton-Jacobi-Bellman (HJB) (Authors,) equation we will use for the dynamic programming.

We can rewrite it by moving $\frac{\partial V}{\partial t}$ to the other side:

$$-\frac{\partial V}{\partial t} = \min_{v_t} \left\{ \eta v_t^2 + \gamma\sigma^2 x_t^2 + \frac{\partial V}{\partial x} v_t \right\} \tag{3}$$

To find the solution to the HJB problem, we first solve the optimal control function that will minimize the Hamiltonian.

We replace the state variable x with q as the inventory size:

$$-\frac{\partial V}{\partial t}(q,t) = \min_{v} \left\{ \eta v_t^2 + \gamma\sigma^2 q^2 + v_t \frac{\partial V}{\partial q}(q,t) \right\}$$

This equation describes the value function $V(q,t)$ for an optimal control problem, where q represents the inventory or position, and v is the control variable representing the trading rate. The goal is to minimize a cost function that includes inventory risk and trading risk. We first find the optimal trading rate, control function v^*, by minimizing the Hamiltonian expression inside the brackets with respect to v.

We do this by taking the partial derivative with respect to v and setting it to zero:

$$\mathcal{L}(v) = \eta v^2 + \gamma\sigma^2 q^2 - v\frac{\partial V}{\partial q}$$

$$\frac{\partial \mathcal{L}}{\partial v} = 2\eta v - \frac{\partial V}{\partial q}$$

Setting the derivative to zero to find the minimum

$$2\eta v - \frac{\partial V}{\partial q} = 0$$

Solving for v:

$$v^* = \frac{1}{2\eta}\frac{\partial V}{\partial q} \tag{4}$$

Substitute the optimal control v^* back into the original HJB equation 3. First, we need the term $\eta(v^*)^2$ and $v^*\frac{\partial V}{\partial q}$

$$\eta(v^*)^2 = \eta\left(\frac{1}{2\eta}\frac{\partial V}{\partial q}\right) = \eta\left(\frac{1}{4\eta^2}\left(\frac{\partial V}{\partial q}\right)^2\right) = \frac{1}{4\eta}\left(\frac{\partial V}{\partial q}\right)^2$$

$$v^*\frac{\partial V}{\partial q} = \frac{1}{2\eta}\left(\frac{\partial V}{\partial q}\right)\frac{\partial V}{\partial q} = \frac{1}{2\eta}\left(\frac{\partial V}{\partial q}\right)^2$$

Substituting these into the HJB equation in 3

$$-\frac{\partial V}{\partial t} = \left[\frac{1}{4\eta}\left(\frac{\partial V}{\partial q}\right)^2 + \gamma\sigma^2 q^2 - \frac{1}{2\eta}\left(\frac{\partial V}{\partial q}\right)^2\right]$$

$$-\frac{\partial V}{\partial t} = \gamma\sigma^2 q^2 - \frac{1}{4\eta}\left(\frac{\partial V}{\partial q}\right)^2$$

We are interested in how the state variable, inventory, changes over time, represented by the PDE $\frac{\partial V}{\partial t}$. By rearranging the terms, we obtain the final form of the solution to the HJB equation in3

$$\frac{\partial V}{\partial t} = \frac{1}{4\eta}\left(\frac{\partial V}{\partial q}\right)^2 - \gamma\sigma^2 q^2 \tag{5}$$

where:

- $V(q,t)$: Value function.
- q: Inventory as the state variable
- $\eta > 0$: Risk aversion coefficient for trading
- $\gamma > 0$: Risk aversion coefficient for inventory
- $\sigma^2 > 0$: Variance of the price process

This is a first order nonlinear partial differential equation for the value function $V(q,t)$. Solving this PDE requires specifying boundary or terminal conditions, such as the value of V at a certain time $t = T$ or for a certain inventory level q.

Simulation

We simulate to find the optimal control by discretizing the inventory and the time bounded limits. We initialize the value function in the terminal state to 0 for all the value paths of the range of the inventory levels. We then iterate backward in time, calculating the state value at each time step.

We use partial derivatives to approximate the finite difference when stepping through the states of the value function. After determining the value of the state function at each time step for the different inventory levels, we calculate and reconstruct the optimal control function to obtain the liquidation size for the next trade.

The value function is the minimum possible cost to go from state q at time t to the terminal time T. The goal is to find $V(q,t)$ for all initial states of q at $t = 0$, and the control policy function that achieves this minimum cost.

We illustrate in Figure (2) the behavior of the optimal liquidation strategy using the cost function in the HJB problem in 3 that is optimized by the control function in 4.

The graph shows how the state value and inventory q change over time based on the optimal trading rate. In a real trading scenario, there are other random price variables that affect the value of the inventory, but for this case the HJB equation in its continuous form optimizes for the expected cost.

The optimal trading rate guides the inventory path towards zero as time approaches the terminal state. A common approach is to assume that the inventory changes discretely on the basis of the liquidation rate.

The slope of $\frac{\partial V}{\partial q}$ indicates how the cost changes as the inventory level increases. The slope is expected to positive for $q > 0$, negative for $q < 0$, and zero at $q = 0$. This means $V(q,t)$ is convex due to the term q^2. The slope with respect to time as it approaches T is negative as the cost is expected to be approaching zero.

The value function represents the expected minimum cost, the equation in (3) indicates that this value is positive.

We refer to the optimal control function in (4) to track the behavior of the optimal trading rate with respect to the inventory as shown in Figure (2)

This plot shows slices of the numerically computed optimal trading rate at different time points. We can observe how the rate decreases as time approaches T (the trading becomes less aggressive when there's less time left, or when the inventory is closer to zero).

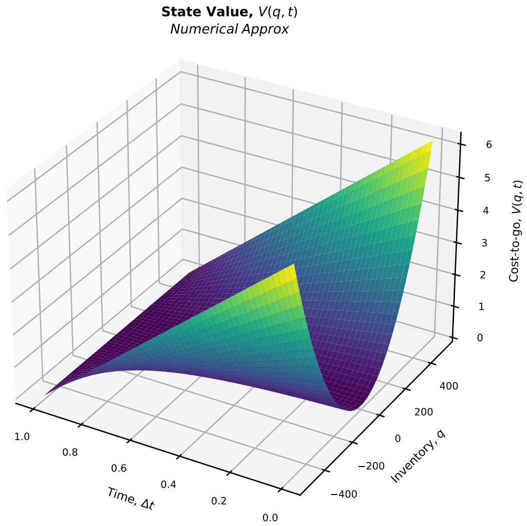

Figure 1: Value Function

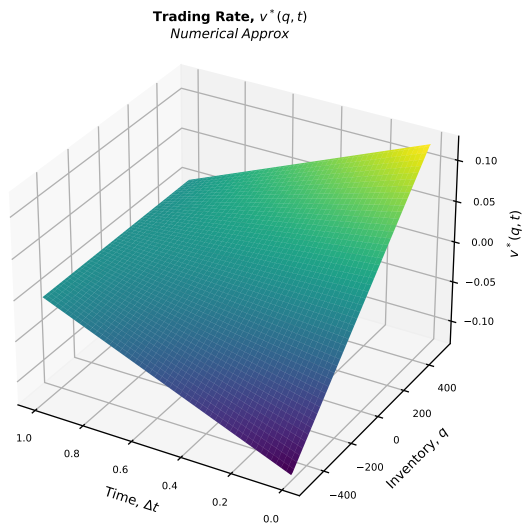

Figure 2: Optimal Control Policy

Analytical Solution

We use the closed form solution from (Almgren & Chriss,) to compare with the numerical approximation simulated earlier.

We define a terminal condition of the value function at the end of the time horizon and the boundary conditions to determine how the value function behaves in certain states.

Boundary Conditions

- **Terminal Condition** Assume that we are trading over a finite time horizon. At the terminal time, the cost is zero. In other words, the value function is zero for all possible inventory levels q.
$$V(q, T) = 0$$
- **Structure of** $V(q, t)$ We assume that the value function has a specific functional form that depends on the state variable and time. A typical guess is a quadratic q and some function of t. As such, we use the ansatz form
$$V(q, t) = -A(t)q^2$$

The negative sign is a common convention in cost minimization problems to ensure that $V(q, t)$ is convex. Here, $A(t)$ is a function of time that we need to determine.

This specific ansatz fits the HJB problem because the underlying system is linear-quadratic (LQ)

- The system dynamics are linear: $dq_t = -\nu_t dt$.
- The running and terminal costs are quadratic: $\gamma\nu^2$, ϕq^2, and αq^2

Substituting $V(t, q) = A(t)q^2$ as the ansatz into the HJB equation, the inventory variable q^2 factors out completely in the equation. This simplifies the complex, multi-variable PDE into a single-variable ODE) for $A(t)$, proving the guess is mathematically correct.

The quadratic ansatz will fail if we introduce constraints that will break the Linear-Quadratic harmony of the problem

- **No-Short-Selling Constraints** Restricting inventory to never drop below zero $q_t \geq 0$ introduces a hard boundary condition.
- **Non-Quadratic Market Impact** If walking the order book scales power-law style (e.g., impact costs proportional to $\nu_t^{1.5}$ instead of ν_t^2, the HJB equation cannot be separated, and will require numerical grid solvers instead of a Riccati equation.

Solve for the Value Function

Now we plug the ansatz into the HJB equation and use the terminal condition to find $A(t)$.

First, determine the partial derivatives of $V(q, t) = -A(t)q^2$:

- Derivative with respect to time:
$$\frac{\partial V}{\partial t} = \frac{\partial}{\partial t}\Big(-A(t)q^2\Big) = -A'(t)q^2$$
- Derivative with respect to inventory:
$$\frac{\partial V}{\partial q} = \frac{\partial}{\partial q}\Big(-A(t)q^2\Big) = -2A(t)q$$

Substitute these into the HJB equation in (5):

$$-A'(t)q^2 = \frac{1}{4\eta}\Big(-2A(t)q\Big)^2 - \gamma\sigma^2 q^2$$

Let's simplify the term with $(\frac{\partial V}{\partial q})^2$:

$$\frac{1}{4\eta}\Big(-2A(t)q\Big)^2 = \frac{1}{4\eta}\Big(4A(t)^2q^2\Big) = \frac{A(t)^2}{\eta}q^2$$

Substitute this into the equation:

$$-A'(t)q^2 = \frac{A(t)^2}{\eta}q^2 - \gamma\sigma^2 q^2$$

Since this equation must hold for all q, we can divide both sides by q^2 (assuming $q \neq 0$):

$$-A'(t) = \frac{A(t)^2}{\eta} - \gamma\sigma^2$$

Rearranging this equation to get a standard form for a first-order ordinary differential equation:

$$A'(t) = \gamma\sigma^2 - \frac{A(t)^2}{\eta}$$

Now, let's apply the terminal condition $V(q, T) = 0$.

Since $V(q, T) = -A(T)q^2$, for this to be zero for all q, we must have $A(T) = 0$.

So, we need to solve the ODE:

$$\frac{dA}{dt} = \gamma\sigma^2 - \frac{A^2}{\eta}$$

with the boundary condition $A(T) = 0$. This is a Riccati equation (Authors,) and to solve we can rearrange it

$$\frac{dA}{dt} = \frac{\gamma\sigma^2\eta - A^2}{\eta}$$

$$\frac{\eta}{\gamma\sigma^2\eta - A^2} dA = dt$$

We can integrate both sides. Let's integrate from t to T on the left and from $A(t)$ to $A(T) = 0$ on the right:

$$\int_{A(t)}^{A(T)} \frac{\eta}{\gamma\sigma^2\eta - A^2} dA = \int_t^T dt$$

The right side is straightforward:

$$\int_t^T dt = T - t$$

For the left side, we use the integral of $\frac{1}{a^2-x^2}$, which is $\frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$. In our case, $a^2 = \gamma\sigma^2\eta$, so $a = \sqrt{\gamma\sigma^2\eta}$. Let $C = \gamma\sigma^2\eta$. The integral is $\int \frac{\eta}{C-A^2} dA$.

$$\int \frac{\eta}{C - A^2} dA = \eta \int \frac{1}{C - A^2} dA = \eta \left(\frac{1}{2\sqrt{C}} \ln \left| \frac{\sqrt{C} + A}{\sqrt{C} - A} \right| \right)$$

$$= \frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A}{\sqrt{\gamma\sigma^2\eta} - A} \right|$$

$$= \frac{\sqrt{\eta}}{2\sqrt{\gamma\sigma^2}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A}{\sqrt{\gamma\sigma^2\eta} - A} \right|$$

$$= \frac{\sqrt{\eta}}{2\sigma\sqrt{\gamma}} \ln \left| \frac{\sigma\sqrt{\gamma\eta} + A}{\sigma\sqrt{\gamma\eta} - A} \right|$$

Applying the limits of integration:

$$\left[\frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A}{\sqrt{\gamma\sigma^2\eta} - A} \right| \right]_{A(t)}^{0} = T - t$$

Substitute $A(T) = 0$:

$$\frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + 0}{\sqrt{\gamma\sigma^2\eta} - 0} \right| - \frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A(t)}{\sqrt{\gamma\sigma^2\eta} - A(t)} \right| = T - t$$

$$\frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln(1) - \frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A(t)}{\sqrt{\gamma\sigma^2\eta} - A(t)} \right| = T - t$$

$$0 - \frac{\eta}{2\sqrt{\gamma\sigma^2\eta}} \ln \left| \frac{\sqrt{\gamma\sigma^2\eta} + A(t)}{\sqrt{\gamma\sigma^2\eta} - A(t)} \right| = T - t$$

Let $K = \sqrt{\gamma\sigma^2\eta}$.

$$-\frac{\eta}{2K} \ln \left| \frac{K + A(t)}{K - A(t)} \right| = T - t$$

$$\ln \left| \frac{K + A(t)}{K - A(t)} \right| = -\frac{2K(T-t)}{\eta}$$

Exponentiate both sides:

$$\left| \frac{K + A(t)}{K - A(t)} \right| = e^{-\frac{2K(T-t)}{\eta}}$$

Since we expect $A(t)$ to be positive for $t < T$ (as V is negative and q^2 is positive, and we are minimizing cost), and $K > 0$, the term $K + A(t)$ will be positive and $K - A(t)$ will be positive (because $A(T) = 0$ and $A'(t) > 0$ if $A < \sqrt{\eta\gamma\sigma^2}$), so we can remove the absolute value.

$$\frac{K + A(t)}{K - A(t)} = e^{-\frac{2K(T-t)}{\eta}}$$

Let $E = e^{-\frac{2K(T-t)}{\eta}}$.

$$K + A(t) = E(K - A(t))$$

$$K + A(t) = EK - EA(t)$$

$$A(t) + EA(t) = EK - K$$

$$A(t)(1 + E) = K(E - 1)$$

$$A(t) = K\frac{E - 1}{E + 1}$$

Substitute $K = \sqrt{\gamma\sigma^2\eta}$ and $E = e^{-\frac{2\sqrt{\gamma\sigma^2\eta}(T-t)}{\eta}}$:

$$A(t) = \sqrt{\gamma\sigma^2\eta}\frac{e^{-\frac{2\sqrt{\gamma\sigma^2\eta}(T-t)}{\eta}} - 1}{e^{-\frac{2\sqrt{\gamma\sigma^2\eta}(T-t)}{\eta}} + 1}$$

We can simplify this expression using the hyperbolic tangent function, $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

Let $x = \frac{\sqrt{\gamma\sigma^2\eta}(T-t)}{\eta}$. Then $E = e^{-2x}$.

$$A(t) = K\frac{e^{-2x} - 1}{e^{-2x} + 1} = K\frac{-(1 - e^{-2x})}{1 + e^{-2x}}$$

Multiply numerator and denominator by e^x:

$$A(t) = K\frac{-(e^x - e^{-x})}{e^x + e^{-x}} = -K\tanh(x)$$

$$A(t) = -\sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right)$$

Recall that $V(q, t) = -A(t)q^2$. Therefore, the solution for the value function is

$$V(q, t) = -\left(-\sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right) \right) q^2$$

$$V(q, t) = \sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right) q^2 \tag{4}$$

Let's check the terminal condition: As $t \to T$, $(T - t) \to 0$, so $\tanh(0) = 0$, which makes $V(q, T) = 0$. This matches our condition. The solution $V(q, t)$ represents the minimum expected cost from time t to time T, given the current inventory level q. The optimal trading strategy v^* can also be found using $v^* = \frac{1}{2\eta}\frac{\partial V}{\partial q}$.

$$\frac{\partial V}{\partial q} = 2q\sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right)$$

$$v^*(q, t) = \frac{1}{2\eta}\left(2q\sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right) \right)$$

$$v^*(q, t) = \frac{q}{\eta}\sqrt{\gamma\sigma^2\eta}\tanh\left(\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{\eta} \right)$$

We simplify $\frac{1}{\eta} \to \sqrt{\frac{1}{\eta^2}}$

$$v^*(q, t) = q\sqrt{\frac{1}{\eta^2}}\sqrt{\gamma\sigma^2\eta}\tanh\left(\sqrt{\frac{1}{\eta^2}}\frac{\sqrt{\gamma\sigma^2\eta}(T - t)}{1} \right)$$

$$v^*(q, t) = q\sqrt{\frac{\gamma\sigma^2\eta}{\eta^2}}\tanh\left(\sqrt{\frac{\gamma\sigma^2\eta}{\eta^2}}(T - t) \right)$$

$$v^*(q,t) = q\sqrt{\frac{\gamma\sigma^2}{\eta}}\tanh\left(\sigma\sqrt{\frac{\gamma}{\eta}}(T-t)\right) \tag{5}$$

The optimal trading rate is proportional to the current inventory q, and depends on the time remaining until horizon T. As time runs out, the tanh term approaches 0, meaning the optimal trading rate slows down, which is intuitive.

We simulate using the the closed form solutions in 4 and 5 by discretizing the inventory and the time to their bounded limits. We initialize the value function at the terminal state to zero. After that, we iterate backward in time in calculating the state value at each time step.

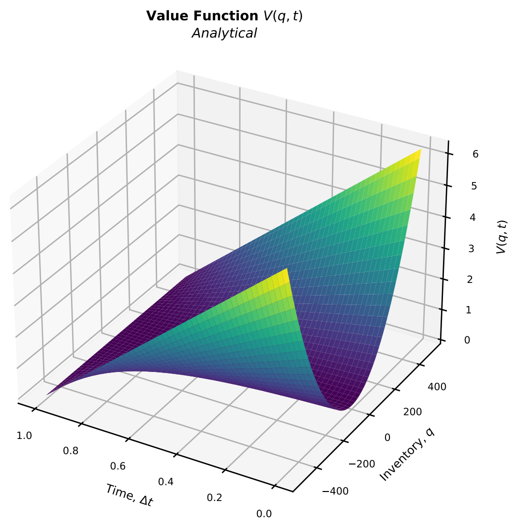

Figure 3: Value Function

The q^2 term in (4) makes the graph convex. The graph will generally show that the $V(q,t)$ is higher for larger absolute inventory levels ($|q|$). This is because holding more inventory (or having a larger net short position) incurs a higher cost/risk, as captured by the $\gamma\sigma^2 q^2$ term in the original cost function.

Looking towards the left side of the plot (where t is close to 0), the value function will be at its highest for any given q. This indicates that the initial state has the highest expected future cost, and the entire purpose of the optimal control strategy is to reduce this cost to zero by time T.

The slope of $\frac{\partial V}{\partial t}$ indicates how the trading rate changes with inventory. The slope of the trading rate and value function towards the terminal time T is negative, since the tanh decreases as time increases and $v^*(q,t)$ becomes smaller. The slope is positive as the inventory increases, corresponding to a higher trading rate when larger inventories are executed.

The graph will generally show that the trading rate is higher for larger absolute inventory levels ($|q|$). This means the trader should trade more aggressively when there are more inventory to liquidate.

Conclusion

We showed how to liquidate large asset positions by simulations which require balancing market impact costs against timing risk through an optimal deterministic trading trajectory. The same model scheme can be used for asset acquisition with minor modification in the model framework used in the liquidation process. The graphs from the analytical solution confirm the value function behavior demonstrated by the Bellman dynamic programming in the numerical approximation. This shows the trade-offs between inventory risk and trading cost in different inventory states at various times.

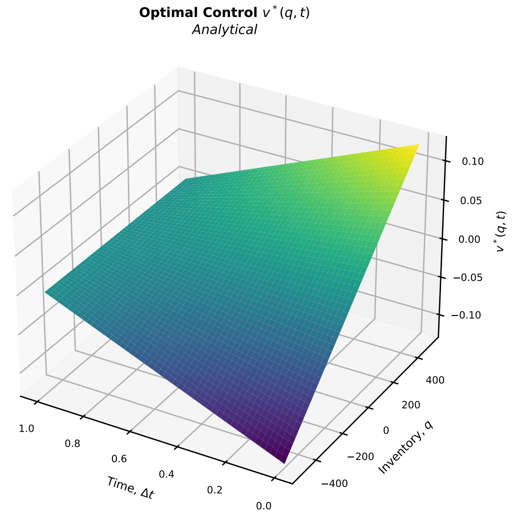

Figure 4: Optimal Control Policy

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